

EEG report classification with MedGemma-27B — summary

Turning free-text clinical EEG reports into five structured diagnostic labels (overall abnormality; focal / generalized epileptiform; focal / generalized non-epileptiform) with a small, CPU-run, grammar-constrained **MedGemma-27B**. Benchmarked against the paper’s **Mistral-7B** (Tian et al.) and a **human** second annotator (SG), on 1994 reports (two neurologists, “Zoe” + “Maria”), scored against reference annotator LD.

Key result: our best prompt, **v5**, reaches **87.6%** whole-report accuracy — it **beats Mistral-7B on all five categories** and comes within ~2 points of the human annotator (89.8%).

Where each version lands

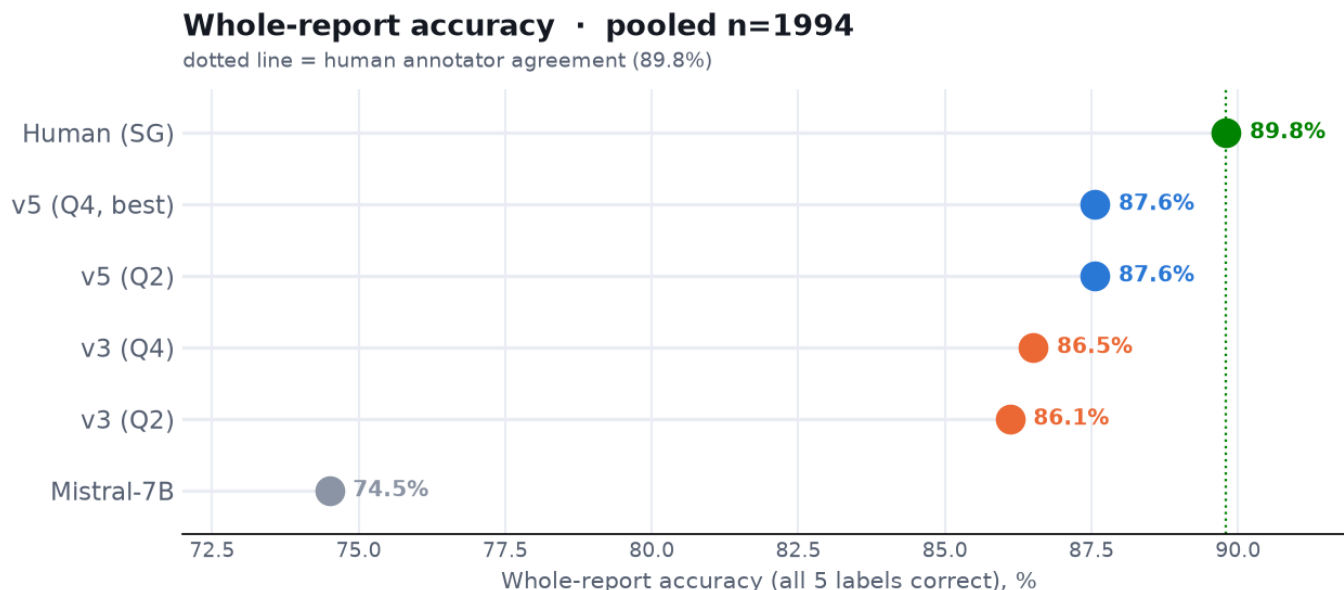


Figure 1: Whole-report accuracy

Two prompt generations, at both model sizes (Q2 ≈ 10 GB, Q4 ≈ 15 GB): **v3** adds an explicit rule for the rare *focal epileptiform* class; **v5** adds a *focal-vs-generalized* rule for slowing. Each step climbs toward the human line; **v5 (Q4)** is the best.

Per category

Each model is a dumbbell: filled dot = **Core F1** (present/absent — number on the right), open dot = **Certainty F1** (exact 1–4 level — number on the left); the line is the drop. On **Core**, our v5 beats Mistral (grey) in every category — by 10–15 points on the harder three — and lands at the human level on the epileptiform categories. On **Certainty**, Mistral’s open circles collapse (Focal Epi 41) while ours hold.

Zoe (in-distribution, n=1495):

Maria (out-of-distribution, n=499):

Numbers (pooled n = 1994, Core F1 vs LD)

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
Mistral-7B	94.7	82.8	74.8	75.6	75.2	74.5
v3 (Q2)	95.2	88.5	90.4	88.2	85.3	86.1
v3 (Q4)	95.9	86.4	89.2	86.9	89.6	86.5

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
v5 (Q2)	95.4	87.6	89.9	88.8	86.7	87.6
v5 (Q4, best)	96.0	85.6	88.8	89.0	89.3	87.6
Human (SG)	98.0	85.7	87.5	90.8	90.0	89.8

Per-category values are F1 (0-100), which measures how well each finding is caught — the standard metric for these imbalanced classes. “Whole report” is the share of reports with all five labels correct.

By dataset — Zoe (in-distribution, n = 1495):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
Mistral-7B	96.1	84.0	72.5	76.0	79.0	75.3
v3 (Q2)	95.9	87.8	91.8	87.9	87.7	86.3
v3 (Q4)	96.4	85.3	90.8	85.5	91.0	86.1
v5 (Q2)	96.1	87.1	91.2	89.0	87.9	87.8
v5 (Q4)	96.3	84.1	90.2	88.1	90.6	87.2
Human (SG)	97.9	83.7	88.7	89.5	89.9	88.8

By dataset — Maria (out-of-distribution, n = 499):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole report
Mistral-7B	89.8	80.6	83.7	74.5	54.0	71.9
v3 (Q2)	92.6	90.0	85.7	89.0	72.1	85.6
v3 (Q4)	94.5	88.9	83.7	90.8	81.6	87.8
v5 (Q2)	93.2	88.5	85.7	88.1	80.0	86.8
v5 (Q4)	95.0	88.9	83.7	91.3	82.2	88.8
Human (SG)	98.1	90.0	83.7	94.4	90.4	92.8

Confidence — Certainty (exact-level) F1

The **Certainty** metric is stricter: the model must also get the exact 1-4 confidence level right. This is where our approach separates most from Mistral — on the per-category chart above it is the **open circle** (the drop from the filled Core dot). The **Whole** column = share of reports with all five exact levels correct.

Pooled (n = 1994):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole
Mistral-7B	72.4	41.4	69.2	44.8	52.2	40.7
v3 (Q2)	70.0	75.4	83.0	67.6	65.0	68.1
v3 (Q4)	75.6	66.3	75.9	60.6	67.1	66.8
v5 (Q2)	65.8	76.8	83.6	59.3	60.1	66.5
v5 (Q4)	74.1	66.7	72.4	63.1	69.0	67.9
Human (SG)	92.0	72.0	84.1	46.9	58.3	65.9

Zoe (n = 1495):

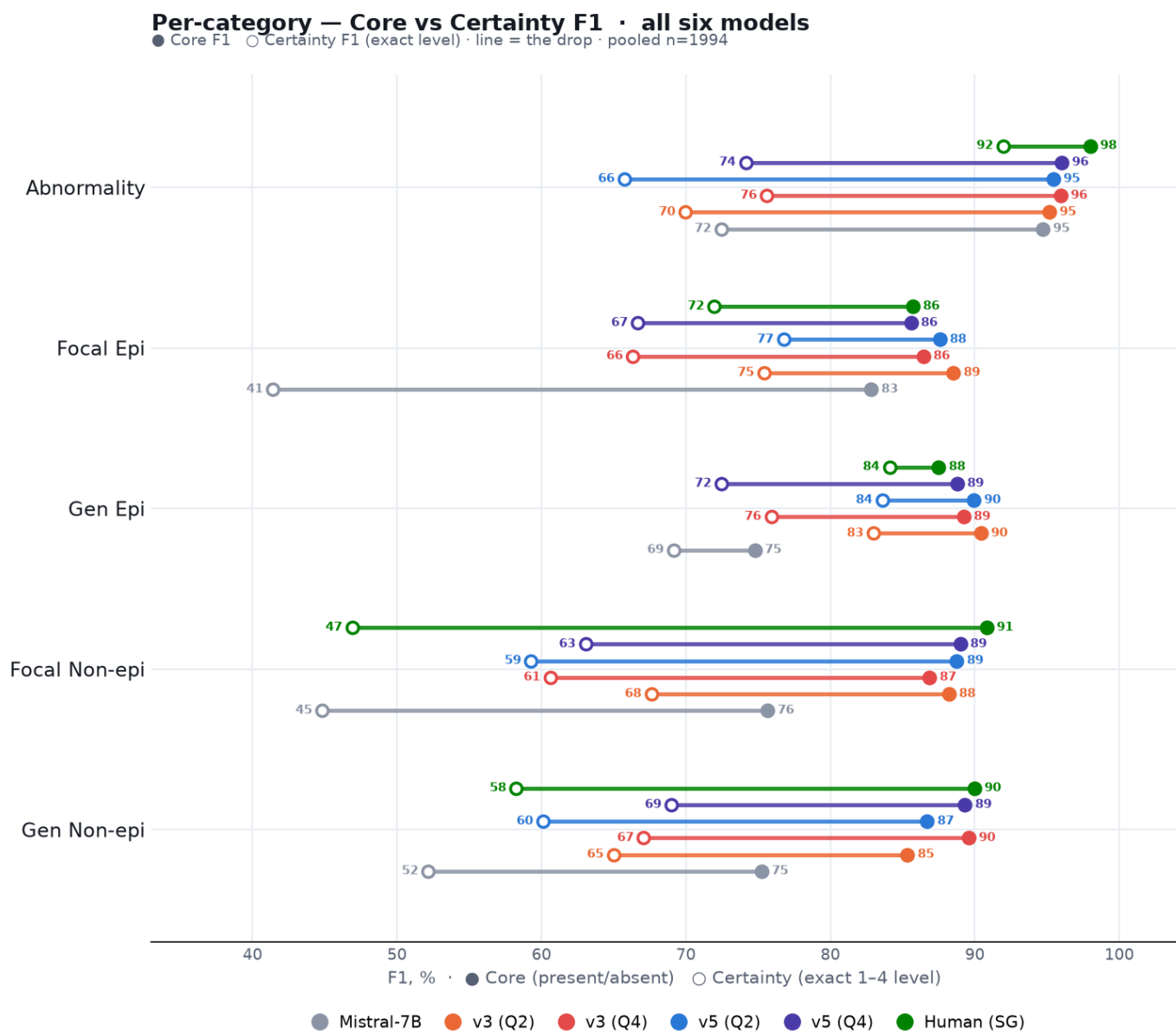


Figure 2: Per-category F1

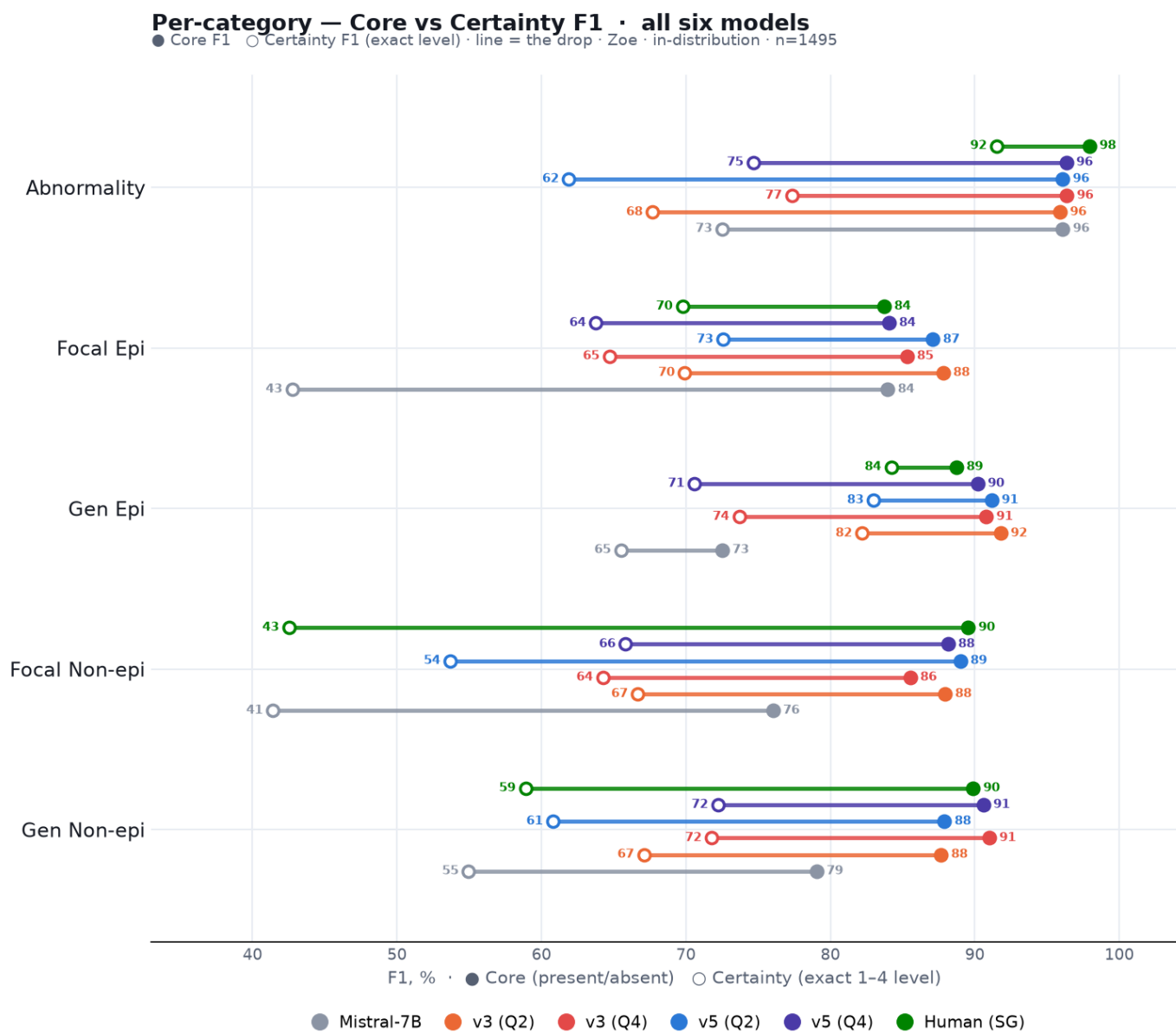


Figure 3: Per-category — Zoe

Per-category — Core vs Certainty F1 · all six models

● Core F1 ○ Certainty F1 (exact level) · line = the drop · Maria · out-of-distribution · n=499

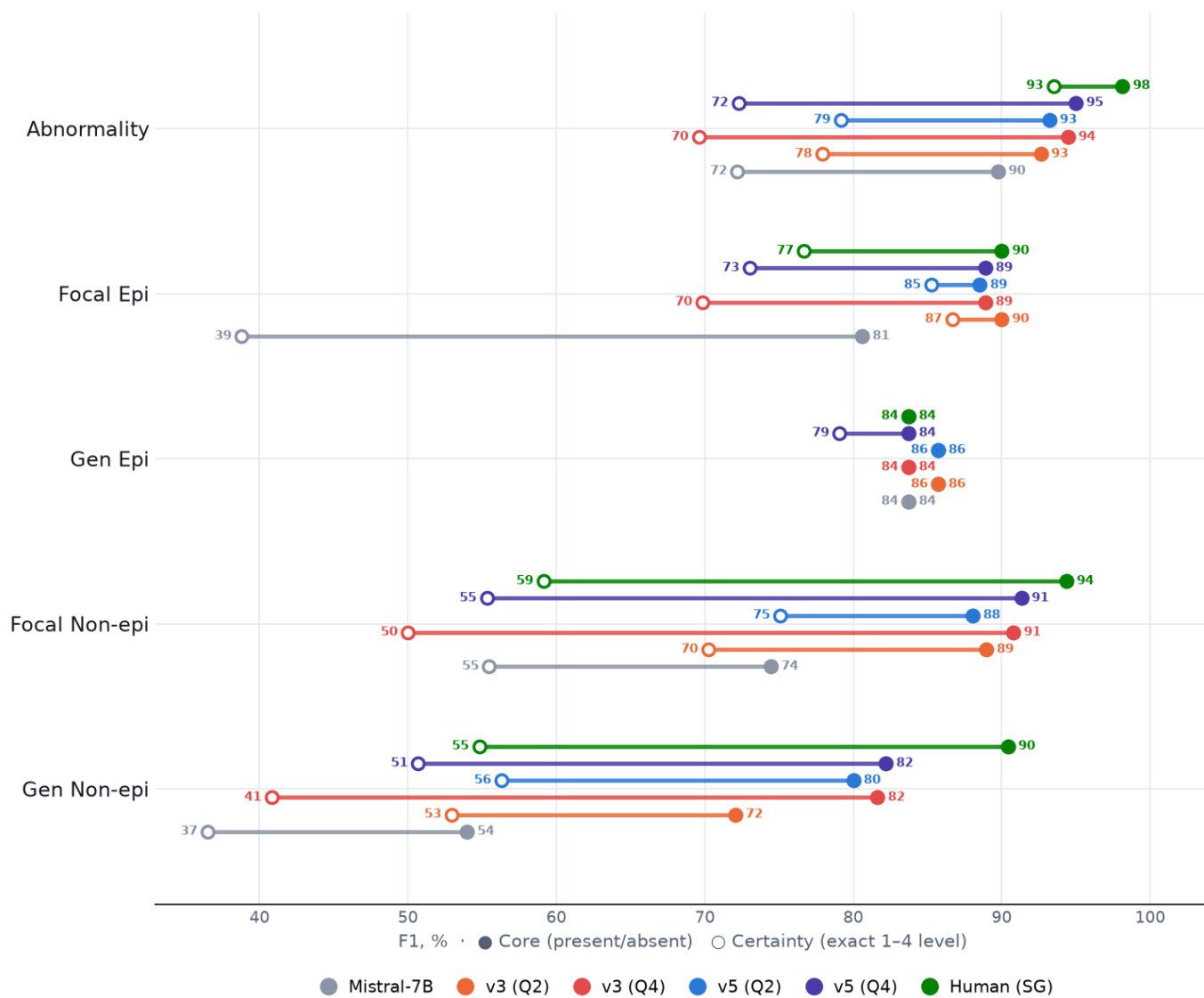


Figure 4: Per-category — Maria

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole
Mistral-7B	72.5	42.7	65.5	41.4	55.0	40.4
v3 (Q2)	67.7	69.9	82.2	66.7	67.1	65.6
v3 (Q4)	77.3	64.7	73.7	64.3	71.8	66.0
v5 (Q2)	61.9	72.6	83.0	53.7	60.8	62.9
v5 (Q4)	74.7	63.8	70.6	65.8	72.2	66.4
Human (SG)	91.5	69.8	84.2	42.5	58.9	63.5

Maria (n = 499):

Model	Abnorm	Focal Epi	Gen Epi	Focal Non	Gen Non	Whole
Mistral-7B	72.2	38.8	83.7	55.5	36.5	41.5
v3 (Q2)	77.9	86.7	85.7	70.2	52.9	75.6
v3 (Q4)	69.6	69.8	83.7	50.0	40.8	69.1
v5 (Q2)	79.2	85.2	85.7	75.1	56.3	77.4
v5 (Q4)	72.3	73.0	79.1	55.3	50.7	72.5
Human (SG)	93.5	76.7	83.7	59.1	54.8	73.3

On focal epileptiform, Mistral drops to ~40 while v3/v5 hold ~65–86. Even the human’s certainty on the slowing classes is modest — those are genuinely hard to pin to an exact level.

Key points

- **Beats the external baseline everywhere.** v5 is above Mistral-7B in all five categories, decisively on the harder ones (e.g. Gen Epi 89 vs 75, Focal Non-epi 89 vs 76).
- **Near-human.** 87.6% vs the 89.8% human ceiling; on the epileptiform classes it already matches the human.
- **Captures confidence, not just the finding.** When the *exact* confidence level must match, Mistral collapses on focal epileptiform (F1 0.41) while v5 holds (0.67–0.77).
- **Generalizes.** The result holds on an unseen neurologist (Maria) and against a held-out second annotator — so it reflects real clinical reasoning, not fitting to one reader’s phrasing.
- **Practical.** Runs on CPU; the smaller Q2 model is enough (same whole-report accuracy as Q4, at $\sim\frac{2}{3}$ the size).

v3 and v5 here are run with a consistency-guaranteeing decoding constraint (a grammar that forbids self-contradictory outputs). Full technical detail, all 10 prompt variants, the tested-and-rejected ideas, and per-dataset tables: [all_prompts.md](#) · prompt texts: [../prompts/](#).